Learning Eigenstructures of Unstructured Data Manifolds

Roy Velich[1] Arkadi Piven[1] David Bensaïd[1] Daniel Cremers[2] Thomas Dagès[1] Ron Kimmel[1]

[1] Technion — Israel Institute of Technology [2] Technical University of Munich

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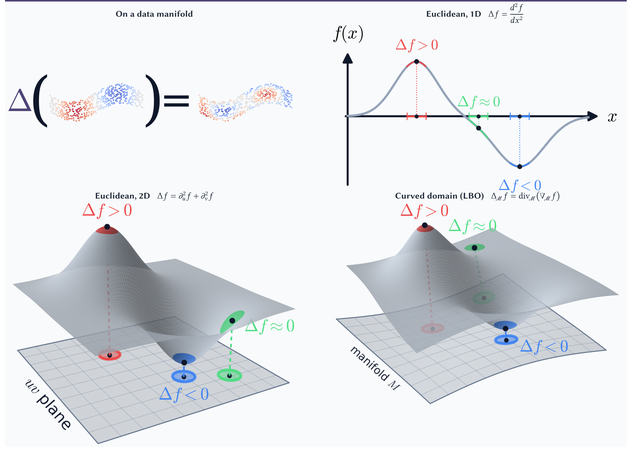
Motivation

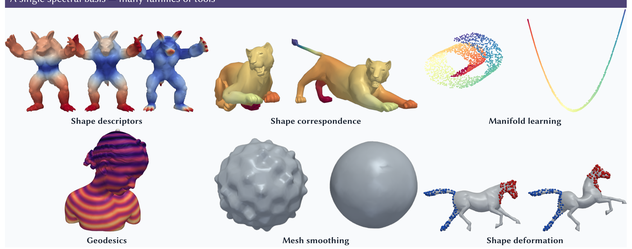
Spectral methods on manifolds underpin shape correspondence, descriptors, manifold learning, ARAP, and many more downstream tools. But computing the Laplacian eigenstructure on raw, unstructured point clouds is delicate: it requires explicit mesh/graph construction, cotangent or heat-kernel weights, then a numerical eigensolve — and none of it is differentiable.

Goal. Learn an ordered eigenbasis of the manifold's Laplace–Beltrami operator (LBO) *directly* from a point sample, bypassing the discretization pipeline.

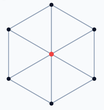
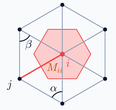
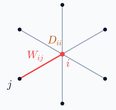
The Laplacian, in one slide



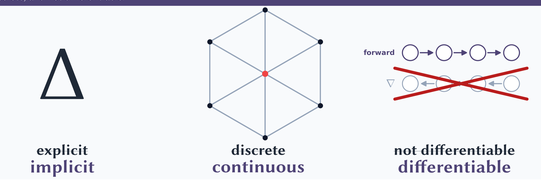
A single spectral basis — many families of tools



Computing the Laplacian eigenstructure, the traditional way

Neighborhood extraction	Operator extraction	Eigensolve	Eigenbasis
		$S\phi = \lambda M \phi$	$\{\phi_i\},\ \{\lambda_i\}$
		$L\,\mathbf{y} = \lambda D\,\mathbf{y}$	$\{\mathbf{y}_i\},\ \{\lambda_i\}$

Explicit, discrete, and not differentiable.



Skip the operator. Learn the eigenbasis directly.

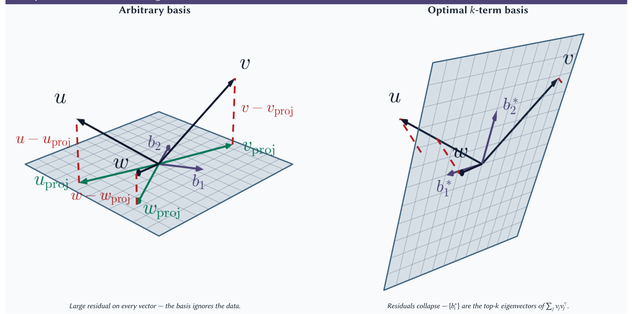
Traditional

Point cloud 〉 Neighborhood extraction 〉 Operator extraction 〉 Eigensolve 〉 **Eigenbasis**

Ours

Point cloud 〉 **Neural network** 〉 **Eigenbasis**

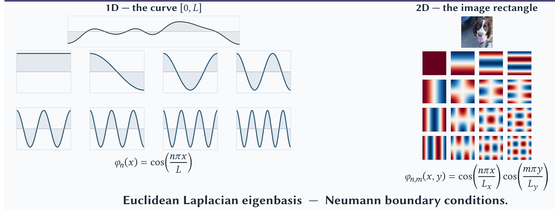
The optimal k-term basis is the eigenbasis



Large residual on every vector — the basis ignores the data.

Residuals collapse — $|b_i^*|$ are the top-k eigenvectors of Σ, $\forall i$.

On a curve or an image, this eigenbasis is the DCT



Euclidean Laplacian eigenbasis — Neumann boundary conditions.

The optimal k-term basis is the eigenbasis of L

Let $L \succeq 0$ be the Laplacian and let

$$\mathscr{C}_L = \{ f \ : \ \langle f, Lf \rangle \le 1 \}, \qquad p_L = \text{Uniform}(\mathscr{C}_L).$$

Theorem 1. *Among all orthonormal k-subspaces, the expected squared reconstruction error under p_L is minimised by the first k eigenvectors of L:*

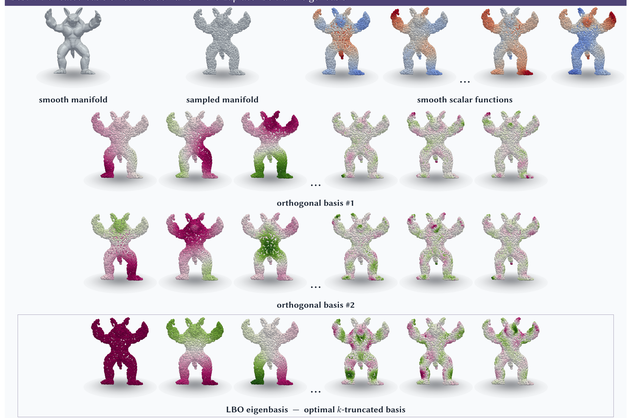
$$\underset{\{\phi_i\}_{i=1}^k}{\operatorname{argmin}} \ \mathrm{E}_{f\sim p_L} \left\| f - \Pi_k f \right\|^2 = \{\varphi_1, \dots, \varphi_k\}.$$

Theorem 2. *Under that basis the worst-case error is sharp:*

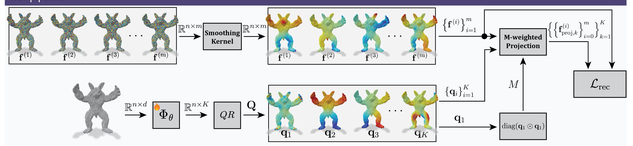
$$\max_{f \in \mathscr{C}_L} \left\| f - \Pi_k f \right\|^2 = \frac{1}{\lambda_{k+1}}.$$

Aflalo, Brezis, Bruckstein, Kimmel & Sochen, C. R. Math. **354** *(2016).*

Best k-truncated basis on curved domains — the Laplace–Beltrami eigenbasis



Our pipeline



Train end-to-end with the average reconstruction loss

$$\mathscr{L}_{\text{rec}} = \frac{1}{mK} \sum_{i=1}^{m} \sum_{k=1}^{K} \| f^{(i)} - f^{(i)}_{\text{proj},k} \|_2^2,$$

measuring how well the first k predicted basis vectors reconstruct each probe $f^{(i)}$, averaged over i and k.

Beyond the LBO

Because the method never builds the operator explicitly, swapping Laplacians is trivial: change only the probe sampler p_L.

Hadamard probes. Substituting random Hadamard probes recovers the same eigenbasis at any prefix.

Schrödinger operator. Adding a potential $V(x)$ to the probe energy yields the Schrödinger eigenstructure $(-\Delta + V)\phi_i = \lambda_i \phi_i$.

figures/schrodinger_grid.pdf

Image manifold

The same recipe applied to CLIP / DINOv2 embeddings of Imagenette and STL10 recovers clusters and outperforms UMAP, t-SNE, PCA, and Laplacian eigenmaps on NMI & ARI.

figures/nmi_ari_bars.pdf